

Question Number	Answer	Additional guidance	Mark
5(a)(i)	The only correct answer is A <i>W</i> <i>B is not correct because X is the phloem</i> <i>C is not correct because Y is the xylem</i> <i>D is not correct because Z is the parenchyma</i>		(1)

Question Number	Answer	Additional guidance	Mark
5(a)(ii)	The only correct answer is B X <i>A is not correct because X is the phloem which transports substances in both directions</i> <i>C is not correct because X is the phloem which transports substances in both directions</i> <i>D is not correct because X is the phloem which transports substances in both directions</i>		(1)

Question Number	Answer	Additional guidance	Mark
5(a)(iii)	The only correct answer is D Y <i>A is not correct because Y is the xylem</i> <i>B is not correct because Y is the xylem</i> <i>C is not correct because Y is the xylem</i>		(1)

Question Number	Answer	Additional guidance	Mark
5(a)(iv)	The only correct answer is D xylem vessels and sclerenchyma fibres <i>A is not correct because phloem sieve tubes do not have secondary thickening</i> <i>B is not correct because phloem sieve tubes do not have secondary thickening</i> <i>C is not correct because sclerenchyma fibres also have secondary thickening</i>		(1)

Question Number	Answer	Additional guidance	Mark
5(b)(i)	An explanation that includes the following points: • magnesium is needed to form chlorophyll / chlorophyll contains magnesium (ions) (1) • for the production of {glucose / carbohydrate / ATP} in photosynthesis (1)	ignore plant growth accept for the conversion of light energy to chemical energy in photosynthesis Accept for one mark ATP / enzymes must be bound to magnesium ion to be biologically active	(2)

Question Number	Answer	Additional guidance	Mark
5(b)(ii)	An answer that includes the following points: • increasing magnesium ion concentration increases {mass / growth} / positive correlation (1) • {same / similar} effect on proportion of mass increase in shoots and roots (1) • significant difference between results as the {SD/range/error} values do not overlap (1) • comment on size of SD linked to reliability of data (1)	accept magnesium deficiency decreases the {mass / growth} accept greater increase of shoot mass than roots accept data is {valid / reliable} as the {SD/range/error} values do not overlap	(4)